CRYPTOLOGY ASSIGNMENT

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1. Write a python code to perform One Time Pad

```
def otp_encrypt(message, key):
    cipher = ""
    for i in range(len(message)):
        m = ord(message[i]) - ord('A')
        k = ord(key[i]) - ord('A')
        value = (m + k) % 26
        cipher += chr(value + ord('A'))
    return cipher
def otp_decrypt(cipher, key):
    message = ""
    for i in range(len(cipher)):
        c = ord(cipher[i]) - ord('A')
        k = ord(key[i]) - ord('A')
        value = (c - k) % 26
        message += chr(value + ord('A'))
    return message
text = input("Enter plaintext: ").upper().replace(" ", "")
key = input("Enter key: ").upper().replace(" ", "")
if len(text) != len(key):
    print("Plaintext and key must have the same length.")
else:
    encrypted = otp_encrypt(text, key)
    print("Encrypted text:", encrypted)
    print("Decrypted text:", otp_decrypt(encrypted, key))
```

2. Write a Python code to break the shift cipher "MCI RWR WH"

```
def shift_decrypt(cipher, key):
    result = ""
    for ch in cipher:
        if ch.isalpha():
            num = ord(ch) - ord('A')
            num = (num - key) % 26
            result += chr(num + ord('A'))
        else:
            result += ch
    return result
cipher_text = "MCI RWR WH"
for key in range(26):
    plain_text = shift_decrypt(cipher_text, key)
    print("Key", key, ":", plain_text)
```

3. Write a Python code to encrypt Rail Fence

```
def rail_fence_encrypt(text, rails):
    rows = [""] * rails
    row = 0
    step = 1
    for ch in text:
        rows[row] += ch
        if row == 0:
            step = 1
        elif row == rails - 1:
            step = -1
        row += step
    return "".join(rows)
```

```
text = input("Enter plaintext: ").replace(" ", "")
rails = int(input("Enter number of rails: "))
if rails < 2 or rails > len(text):
    print("Invalid number of rails")
else:
    print("Encrypted text:", rail_fence_encrypt(text, rails))
```